

**MANIPAL UNIVERSITY JAIPUR**

School of Computer Science & Engineering

Department of Artificial Intelligence and Machine learning

Machine Learning | AIM3101 | Credits 3 1 0 4

**Assignment 1**

**Que 1:** A model gives the following results:

| Model             | Training MSE | Validation MSE |
|-------------------|--------------|----------------|
| Linear Regression | 8.2          | 25.6           |
| Ridge Regression  | 10.1         | 16.8           |
| Lasso Regression  | 10.8         | 15.2           |

Which model would you select for deployment? Do not select the model based only on training error. Justify your answer using the concepts of overfitting, regularization and generalization.

**Que 2:** A machine-learning engineer trains three models on the same dataset to predict house prices. The following results are obtained:

| Model   | Training Error | Validation Error |
|---------|----------------|------------------|
| Model A | 2%             | 18%              |
| Model B | 8%             | 10%              |
| Model C | 15%            | 17%              |

Answer the following:

- Identify which model is suffering primarily from high variance, which is suffering from high bias, and which provides the best bias–variance trade-off. Justify your answer.
- Explain how the difference between training and validation error can be used to diagnose overfitting and underfitting.
- For Model A, suggest two techniques that could reduce variance without substantially increasing bias.

(d) Suppose increasing the model complexity of Model C reduces its training error from 15% to 5%, but its validation error increases from 17% to 22%. Explain what this indicates about the relationship between model complexity and variance.

**Que 3:** Derive the Naïve Bayes classification rule starting from Bayes' theorem. Clearly explain how the assumption of conditional independence among features simplifies the computation.

**Que 4:** A classifier has a high ROC-AUC but poor Precision at the operating threshold used by an organization. Explain how this situation can occur and why AUC alone should not determine the final classification threshold.

**Que 5:** If you train a linear SVM on a dataset of 10,000 points and obtain 5 support vectors, what happens to the decision boundary if you remove the other 9,995 points and retrain the model? What happens if you remove one of the 5 support vectors?

**Que 6:** The C parameter in trade-off, explain how changing the regularization parameter C affects the bias-variance trade-off. What happens to the margin width and the number of support vectors as  $(C \rightarrow \infty)$ ?